

Numerical studies on a novel split-step quadratic B -spline finite element method for the coupled Schrödinger–KdV equations

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ABSTRACT

We propose a novel split-step quadratic B -spline finite element method for solving the initial-boundary value problem of the coupled Schrödinger–KdV equations. A full-discrete finite element scheme is constructed. The conserved properties of the full-discrete scheme are proved. Detailed numerical results show the efficiency of our scheme.

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1. Introduction

The coupled nonlinear Schrödinger–KdV (NLS–KdV) equations have attracted extensive interest in physics and mathematics [1–9]. Numerical simulations of the propagation and interactions of one-dimensional Langmuir solitons and their generation from random fluctuations by an external pump field are presented in [1]. Results on existence, uniqueness of the solutions and global attractor of this systems have been obtained in [2,3]. The numerical methods are important tools to understand the feature of the NLS–KdV equations and to obtain possible physical applications. Even though numerous computational methods have been investigated for the single Schrödinger equation or KdV equation, very limited analytic ways for modelling the coupled system are available because of its complexity. Appert and Vaclavik [4] did some numerical work concerning this NLS–KdV equations by the finite difference Crank–Nicolson (CN) scheme. In our previous works [10], the finite element method (FEM) was successfully used to study the initial-boundary value problem for the NLS–KdV equations.

Recently several authors (e.g. [11,12]) proposed a split-step method for solving Schrödinger equation with various geometries. The main idea of operator splitting methods is to decompose the considered problem into linear and nonlinear subproblems on each time step. Then the subproblems can be well settled separately. Wang [13] proposed some numerical studies for various version nonlinear Schrödinger equations using the operator splitting finite difference method. Meanwhile, the B -spline functions provide a reasonable and consistent basis, and can offer superior accuracy for the construction of a large variety of numerical problems [14–20]. However, the numerical studies of the NLS–KdV equations by a split-step B -spline finite element method have been paid less attention.

In this paper, we propose a split-step quadratic B -spline finite element method (SSQBS FEM) for solving the NLS–KdV equations, which is an extension of the previous work [10]. For the nonlinear Schrödinger-type equation, we split it into a linear part and a nonlinear part. The solution of the nonlinear one can be solved exactly and then be used as the initial

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value of the linear one. The linear one and the KdV equation are solved by the FEM. Meanwhile, a new variable is introduced to substitute the nonlinear parts in KdV equation, which can linearize the nonlinear terms and save mostly computing time [21]. Moreover, the CN time difference discrete is applied to construct the full-discrete scheme. The conserved properties of this scheme are presented. Numerical results are consistency well with the theoretical results.

This paper is organized as follows. In Section 2, we introduce the initial-boundary value problem about the NLS–KdV equations and some preliminaries. Section 3 is devoted to describe the detailed process of constructing the full-discrete scheme and computing steps. The conserved properties of the discrete scheme are presented in Section 4. In Section 5, numerical experiments and data comparison are provided to verify the accuracy and efficiency. The last section is conclusion.

2. Initial-boundary value problem and some preliminaries

The NLS–KdV equations describing the nonlinear dynamics of one-dimensional Langmuir and ion-acoustic waves in a system of coordinates moving at the ion-acoustic speed have the form [4–6]:

$$i\epsilon u_t + \frac{3}{2}u_{xx} - \frac{1}{2}uv = 0, \quad (x, t) \in (x_l, x_r) \times (0, T], \quad (2.1)$$

$$v_t + \frac{1}{2}v_{xxx} + \frac{1}{2}(|u|^2 + v^2)_x = 0, \quad (x, t) \in (x_l, x_r) \times (0, T], \quad (2.2)$$

$$u(x_l, t) = u(x_r, t) = 0, \quad v(x_l, t) = v(x_r, t) = 0, \quad t \in (0, T], \quad (2.3)$$

$$u(x, 0) = u_0(x), \quad v(x, 0) = v_0(x), \quad x \in (x_l, x_r), \quad (2.4)$$

where the complex-valued function u and the real-valued function v represent the electric field of Langmuir oscillations and the low-frequency density perturbation, respectively. $\epsilon > 0$ is a constant. $u_0(x)$ and $v_0(x)$ are known smooth functions and satisfy

$$u_0(x_l) = u_0(x_r) = 0, \quad v_0(x_l) = v_0(x_r) = 0. \quad (2.5)$$

Let $\Omega = [x_l, x_r]$. We use the notation $H^r(\Omega) = W^{r,2}(\Omega)$ to denote the complex Sobolev space on Ω .

$$H_0^r = \left\{ u \in H^r(\Omega) \mid \partial_x^j u(x_l) / \partial x^j = \partial_x^j u(x_r) / \partial x^j = 0, \quad j = 0, 1, \dots, r-1 \right\},$$

where $\partial_x^j u / \partial x^j$ represent the j th derivative with respect to x of function u .

Denote $(\cdot, \cdot)$ the inner product in $L_2(\Omega)$, $\|\cdot\|$ the norm in $L_2(\Omega)$ and $\|\cdot\|_r$ the norm in space $H^r(\Omega)$.

$$(u, v) = \int_{\Omega} u \cdot \bar{v} dx, \quad \|v\| = (v, v)^{\frac{1}{2}} = \left(\int_{\Omega} v \cdot \bar{v} dx \right)^{\frac{1}{2}}, \quad \|v\|_r = \|v\|_{H^r} = \left(\sum_{|x| \leq r} \|D^x v\|^2 \right)^{\frac{1}{2}},$$

where the $\bar{v}$ denotes the conjugate of v .

The initial-boundary value problem for (2.1)–(2.4) has the following invariants:

the number of plasmons:

$$I_1 = \int_{\Omega} |u|^2 dx, \quad (2.6a)$$

the number of particles:

$$I_2 = \int_{\Omega} v dx, \quad (2.6b)$$

the energy of the oscillations:

$$I_3 = \int_{\Omega} [3|u_x|^2 + v|u|^2 + \frac{1}{3}v^3 - \frac{1}{2}(v_x)^2] dx, \quad (2.6c)$$

which play an important role in the analysis of this equations.

3. Numerical method

3.1. Weak form

First, we divide the time interval $[0, T]$ into N subintervals of length $\tau = T/N$, where N is a positive integer. Denote $t_n = n\tau$ and $t_{n+\frac{1}{2}} = (n + \frac{1}{2})\tau$, $n = 0, 1, \dots, N$, respectively. From t_n to t_{n+1} , the Eq. (2.1) is split into two subequations:

$$u_t = -\frac{i}{2\epsilon}uv, \quad (3.1)$$

$$u_t = \frac{3i}{2\epsilon}u_{xx}. \quad (3.2)$$

The Eq. (3.1) can be solved exactly by applying the value of the n th level as initial value. Using the solution of (3.1) as the initial value of (3.2), we solve (3.2) by applying the FEM to get the value of $(n+1)$ th level.

Next, let $\phi = |u|^2 + v^2$. The Eq. (2.2) is transformed into the following form:

$$\phi = |u|^2 + v^2, \quad (3.3)$$

$$v_t + \frac{1}{2} v_{xxx} + \frac{1}{2} \phi_x = 0, \quad (3.4)$$

with the boundary condition $u(x_t, t) = u(x_r, t) = 0, \phi(x_t, t) = \phi(x_r, t) = 0$, and initial condition $u(x, 0) = u_0(x), \phi(x, 0) = |u_0(x)|^2 + (v_0(x))^2$.

Applying the Galerkin method to (3.2)–(3.4) with test function $\varphi(x) \in H_0^1$, integrating by parts and using the boundary conditions yield the weak form:

$$(u_t, \varphi) = -\frac{3i}{2\epsilon} (u_x, \varphi_x), \quad (3.5)$$

$$(\phi, \varphi) = (|u|^2 + v^2, \varphi), \quad (3.6)$$

$$(v_t, \varphi) - \frac{1}{2} (v_{xx}, \varphi_x) - \frac{1}{2} (\phi, \varphi_x) = 0, \quad (3.7)$$

$$(u(x, 0), \varphi) = (u_0(x), \varphi), (v(x, 0), \varphi) = (v_0(x), \varphi), \quad (3.8)$$

$$(\phi(x, 0), \varphi) = (|u_0(x)|^2 + (v_0(x))^2, \varphi). \quad (3.9)$$

3.2. Full-discrete scheme

For the sake of the implementation of the FEM, the finite interval $[x_l, x_r]$ is partitioned into M finite elements of equal length h by the nodes x_i ($i = 0, 1, 2, \dots, M$) such that $x_l = x_0 < x_1 < x_2 < \dots < x_M = x_r$, and $h = \Delta x = x_i - x_{i-1} = (x_r - x_l)/M$. Define $S_h(\Omega) \subset H_0^1(\Omega)$ the piecewise polynomial function space on Ω . $S_h(\Omega)$ possesses the well-known approximation properties [22,23]:

If $u \in H^r(\Omega) \cap H_0^1(\Omega)$, then

$$\inf_{\chi \in S_h} \{ \|u - \chi\| + h \|u - \chi\|_1 \} \leq Ch^s \|u\|_s, \quad 1 \leq s \leq r \quad (3.10)$$

and, for all $\chi \in S_h$, the following inverse inequality holds:

$$\|\chi\|_\beta \leq Ch^{-(\beta-\alpha)} \|\chi\|_\alpha, \quad 0 \leq \alpha \leq \beta \leq r-1, \quad (3.11)$$

where C is independent of h .

Denote $u^n = u(x, t_n), v^n = v(x, t_n), \phi^{n-1/2} = \phi(x, t_{n-1/2})$ and let u_h^n, v_h^n and $\phi_h^{n-1/2} \in S_h(\Omega)$ be the approximation of the functions u^n, v^n and $\phi^{n-1/2}$, respectively. Denote $I_h: C(\Omega) \rightarrow S_h(\Omega)$ an interpolation operator. For conveniently denoting, we define $u^{n*} = \exp(-\frac{i\tau}{2\epsilon} v^n) u^n$ and u_h^{n*} the approximation of u^{n*} which satisfies $u_h^{n*} = \exp(-\frac{i\tau}{2\epsilon} v_h^n) u_h^n$.

On the direction of temporal, the CN difference discrete is used. The full-discrete scheme is defined as: to find u_h^{n+1}, v_h^{n+1} and $\phi_h^{n+1/2}$ belong to $S_h(\Omega)$, for all $\varphi \in S_h(\Omega)$, such that

$$u_h^{n*} = \exp\left(-\frac{i\tau}{2\epsilon} v_h^n\right) u_h^n, \quad (3.12)$$

$$\frac{1}{\tau} (u_h^{n+1} - u_h^{n*}, \varphi) = -\frac{3i}{4\epsilon} (u_h^{n*} + u_h^{n+1}, \varphi_x), \quad (3.13)$$

$$\frac{1}{2} (\phi_h^{n+1/2} + \phi_h^{n-1/2}, \varphi) = (I_h(|u_h^n|^2 + (v_h^n)^2), \varphi), \quad (3.14)$$

$$\frac{1}{\tau} (v_h^{n+1} - v_h^n, \varphi) - \frac{1}{4} (v_h^{n+1} + v_h^n, \varphi_{xxx}) - \frac{1}{2} (\phi_h^{n+1/2}, \varphi_x) = 0, \quad (3.15)$$

$$(u_h^0(x), \varphi) = (u_0(x), \varphi), (v_h^0(x), \varphi) = (v_0(x), \varphi), (\phi_h^{1/2}, \varphi) = (\phi(x, 0), \varphi). \quad (3.16)$$

Selection of the quadratic B -splines (see Fig. 1) as interpolation functions in the FEM satisfies the continuity conditions which is needed in the integral Eqs. (3.13)–(3.16). The quadratic B -splines $\varphi_i (i = -1, 0, 1, 2, \dots, M)$ over the interval $[x_l, x_r]$ are defined by the relationships

$$\varphi_i(x) = \frac{1}{h^2} \begin{cases} (x_{i+2} - x)^2 - 3(x_{i+1} - x)^2 + 3(x_i - x)^2, & [x_{i-1}, x_i], \\ (x_{i+2} - x)^2 - 3(x_{i+1} - x)^2, & [x_i, x_{i+1}], \\ (x_{i+2} - x)^2, & [x_{i+1}, x_{i+2}], \\ 0, & \text{otherwise.} \end{cases} \quad (3.17)$$

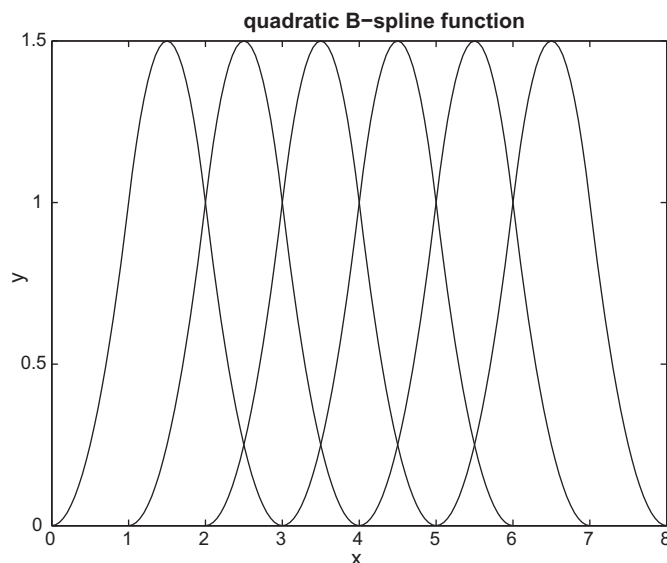


Fig. 1. The figure of the quadratic B-spline function.

The quadratic spline $\varphi_i(x)$ and its first derivative vanish outside the interval $[x_{i-1}, x_{i+2}]$. These properties are required for the following deduction of the discrete scheme. According to the feature of Galerkin FEM, the approximate solutions can be written in the terms of the linear combination of the quadratic splines functions:

$$\begin{aligned} u_h^n &= \sum_{i=-1}^M U_i^n \varphi_i(x), \quad v_h^n = \sum_{i=-1}^M V_i^n \varphi_i(x), \\ \phi_h^{n-1/2} &= \sum_{i=-1}^M \Phi_i^{n-1/2} \varphi_i(x), \quad u_h^{n*} = \sum_{i=-1}^M U_i^{n*} \varphi_i(x), \end{aligned} \quad (3.18)$$

where U_i^n, V_i^n, U_i^{n*} and $\Phi_i^{n-1/2}$ are undetermined coefficients.

The nodal values of u_h^n, v_h^n and $\phi_h^{n-1/2}$ and their first derivative at the knot x_i can be expressed as

$$\begin{aligned} u_h^n(x_i) &= U_{i-1}^n + U_i^n, \quad u_{hx}^n(x_i) = \frac{2}{h}(U_i^n - U_{i-1}^n), \\ v_h^n(x_i) &= V_{i-1}^n + V_i^n, \quad v_{hx}^n(x_i) = \frac{2}{h}(V_i^n - V_{i-1}^n), \\ \phi_h^{n-1/2}(x_i) &= \Phi_{i-1}^{n-1/2} + \Phi_i^{n-1/2}. \end{aligned} \quad (3.19)$$

Considering the approximate solutions must satisfy the boundary conditions (2.3), thus we have

$$u_h^n(x_0) = u_h^n(x_M) = 0, \quad v_h^n(x_0) = v_h^n(x_M) = 0, \quad \phi_h^{n-1/2}(x_0) = \phi_h^{n-1/2}(x_M) = 0, \quad (3.20)$$

or

$$\begin{aligned} U_{-1}^n &= -U_0^n, \quad V_{-1}^n = -V_0^n, \quad \Phi_{-1}^{n-1/2} = -\Phi_0^{n-1/2}, \\ U_{M-1}^n &= -U_M^n, \quad V_{M-1}^n = -V_M^n, \quad \Phi_{M-1}^{n-1/2} = -\Phi_M^{n-1/2}. \end{aligned} \quad (3.21)$$

Define the interpolation operator I_h with the following form:

$$I_h f(u_h^n) = \sum_{i=-1}^M f(U_i^n) \varphi_i(x), \quad (3.22)$$

where

$$f(x) = \begin{cases} |x|^2, & \text{if } x \text{ is complex,} \\ x^2, & \text{if } x \text{ is real.} \end{cases}$$

From (3.17)–(3.22) and the property $(I_h(f(y(x_i)))) = f(y(x_i))$ of the interpolation operator I_h , we get that

$$\begin{aligned} f(U_{-1}^n) &= -f(U_0^n), \quad f(U_{M-1}^n) = -f(U_M^n), \\ f(V_{-1}^n) &= -f(V_0^n), \quad f(V_{M-1}^n) = -f(V_M^n). \end{aligned} \quad (3.23)$$

Actually, the basis functions can be constructed as follows:

$$\begin{aligned} \Psi_0(x) &= \varphi_0(x) - \varphi_{-1}(x), \\ \Psi_i(x) &= \varphi_i(x), \quad i = 1, 2, \dots, M-2, \\ \Psi_{M-1}(x) &= \varphi_{M-1}(x) - \varphi_M(x). \end{aligned} \quad (3.24)$$

Hence we have the approximate solution

$$\begin{aligned} u_h^n &= \sum_{i=0}^{M-1} U_i^n \Psi_i(x), \quad v_h^n = \sum_{i=0}^{M-1} V_i^n \Psi_i(x), \\ \phi_h^{n-1/2} &= \sum_{i=0}^{M-1} \Phi_i^{n-1/2} \Psi_i(x), \quad u_h^{n*} = \sum_{i=0}^{M-1} U_i^{n*} \varphi_i(x). \end{aligned} \quad (3.25)$$

Since all other splines are zeros over interval $[x_i, x_{i+1}]$ except $\Psi_{i-1}(x)$, $\Psi_i(x)$ and $\Psi_{i+1}(x)$, the approximate solutions over this interval have the following forms

$$\begin{aligned} u_h^n &= \sum_{j=i-1}^{i+1} U_j^n \Psi_j(x), \quad \phi_h^{n-1/2} = \sum_{j=i-1}^{i+1} \Phi_j^{n-1/2} \Psi_j(x), \\ v_h^n &= \sum_{j=i-1}^{i+1} V_j^n \Psi_j(x), \quad I_h(f(u_h^n)) = \sum_{j=i-1}^{i+1} f(U_j^n) \Psi_j(x), \\ I_h(f(v_h^n)) &= \sum_{j=i-1}^{i+1} f(V_j^n) \Psi_j(x), \quad u_h^{n*} = \sum_{j=i-1}^{i+1} U_j^{n*} \Psi_j(x). \end{aligned} \quad (3.26)$$

Substituting (3.26) into (3.13)–(3.16) and taking the test function $\varphi(x) = \Psi_k(x)$, $k = 0, 1, 2, \dots, M-1$, we have the discrete scheme over the interval $[x_i, x_{i+1}]$ as following

$$\sum_{j=i-1}^{i+1} U_j^{n+1} (\Psi_j(x), \Psi_k(x)) = -\frac{3i\tau}{4\epsilon} \left[\sum_{j=i-1}^{i+1} U_j^{n*} (\Psi_{jk}(x), \Psi_{kx}(x)) + \sum_{j=i-1}^{i+1} U_j^{n+1} (\Psi_{jk}(x), \Psi_{kx}(x)) \right] + \sum_{j=i-1}^{i+1} U_j^{n*} (\Psi_j(x), \Psi_k(x)), \quad (3.27)$$

$$\sum_{j=i-1}^{i+1} \Phi_j^{n+1/2} (\Psi_j(x), \Psi_k(x)) = 2 \sum_{j=i-1}^{i+1} f(U_j^n) (\Psi_j(x), \Psi_k(x)) + 2 \sum_{j=i-1}^{i+1} f(V_j^n) (\Psi_j(x), \Psi_k(x)) - \sum_{j=i-1}^{i+1} \Phi_j^{n-1/2} (\Psi_j(x), \Psi_k(x)), \quad (3.28)$$

$$\begin{aligned} \sum_{j=i-1}^{i+1} V_j^{n+1} (\Psi_j(x), \Psi_k(x)) - \frac{\tau}{4} \sum_{j=i-1}^{i+1} V_j^{n+1} (\Psi_{jxx}(x), \Psi_{kx}(x)) &= \frac{\tau}{4} \sum_{j=i-1}^{i+1} V_j^n (\Psi_{jxx}(x), \Psi_{kx}(x)) + \frac{\tau}{2} \sum_{j=i-1}^{i+1} \Phi_j^{n+1/2} (\Psi_j(x), \Psi_{kx}(x)) \\ &+ \sum_{j=i-1}^{i+1} V_j^n (\Psi_j(x), \Psi_k(x)). \end{aligned} \quad (3.29)$$

According to the properties of the splines, the subscripts k in $\Psi_k(x)$ can only be taken as $k = i-1, i, i+1$ to make the inner product in (3.27)–(3.29) nonzero.

Denote the i th element $e_i = [x_i, x_{i+1}]$. On this element, the discrete scheme (3.27)–(3.29) can be simplified to matrix notation

$$A^{e_i} U^{(e_i)n+1} + \frac{3i\tau}{4\epsilon} B^{e_i} U^{(e_i)n+1} = -\frac{3i\tau}{4\epsilon} B^{e_i} U^{(e_i)n*} + A^{e_i} U^{(e_i)n*}, \quad (3.30)$$

$$A^{e_i} [\Phi^{(e_i)n+1/2} + \Phi^{(e_i)n-1/2} - 2f(U^{(e_i)n}) - 2f(V^{(e_i)n})] = 0, \quad (3.31)$$

$$A^{e_i} V^{(e_i)n+1} - \frac{\tau}{4} C^{e_i} V^{(e_i)n+1} = \frac{\tau}{4} C^{e_i} V^{(e_i)n} + \frac{\tau}{2} D^{e_i} \Phi^{(e_i)n+1/2} + A^{e_i} V^{(e_i)n}, \quad (3.32)$$

where $U^{(e_i)n}$, $V^{(e_i)n}$, $U^{(e_i)n*}$, $\Phi^{(e_i)n-1/2}$, $f(U^{(e_i)n})$ and $f(V^{(e_i)n})$ are vectors on element e_i ,

$$\begin{aligned} U^{(e_i)n} &= (U_{i-1}^n, U_i^n, U_{i+1}^n)^\top, \quad V^{(e_i)n} = (V_{i-1}^n, V_i^n, V_{i+1}^n)^\top, \\ U^{(e_i)n*} &= (U_{i-1}^{n*}, U_i^{n*}, U_{i+1}^{n*})^\top, \quad \Phi^{(e_i)n-1/2} = (\Phi_{i-1}^{n-1/2}, \Phi_i^{n-1/2}, \Phi_{i+1}^{n-1/2})^\top, \\ f(U^{(e_i)n}) &= (f(U_{i-1}^n), f(U_i^n), f(U_{i+1}^n))^\top, \quad f(V^{(e_i)n}) = (f(V_{i-1}^n), f(V_i^n), f(V_{i+1}^n))^\top. \end{aligned} \quad (3.33)$$

A^{e_i} , B^{e_i} , C^{e_i} , D^{e_i} are 3×3 matrices which have the following forms:

$$\begin{aligned}
A^{e_i} &= \begin{pmatrix} (\Psi_{i-1}(x), \Psi_{i-1}(x)) & (\Psi_i(x), \Psi_{i-1}(x)) & (\Psi_{i+1}(x), \Psi_{i-1}(x)) \\ (\Psi_{i-1}(x), \Psi_i(x)) & (\Psi_i(x), \Psi_i(x)) & (\Psi_{i+1}(x), \Psi_i(x)) \\ (\Psi_{i-1}(x), \Psi_{i+1}(x)) & (\Psi_i(x), \Psi_{i+1}(x)) & (\Psi_{i+1}(x), \Psi_{i+1}(x)) \end{pmatrix}, \\
B^{e_i} &= \begin{pmatrix} (\Psi_{i-1,x}(x), \Psi_{i-1,x}(x)) & (\Psi_{i,x}(x), \Psi_{i-1,x}(x)) & (\Psi_{i+1,x}(x), \Psi_{i-1,x}(x)) \\ (\Psi_{i-1,x}(x), \Psi_{i,x}(x)) & (\Psi_{i,x}(x), \Psi_{i,x}(x)) & (\Psi_{i+1,x}(x), \Psi_{i,x}(x)) \\ (\Psi_{i-1,x}(x), \Psi_{i+1,x}(x)) & (\Psi_{i,x}(x), \Psi_{i+1,x}(x)) & (\Psi_{i+1,x}(x), \Psi_{i+1,x}(x)) \end{pmatrix}, \\
C^{e_i} &= \begin{pmatrix} (\Psi_{i-1,xx}(x), \Psi_{i-1,xx}(x)) & (\Psi_{i,xx}(x), \Psi_{i-1,xx}(x)) & (\Psi_{i+1,xx}(x), \Psi_{i-1,xx}(x)) \\ (\Psi_{i-1,xx}(x), \Psi_{i,xx}(x)) & (\Psi_{i,xx}(x), \Psi_{i,xx}(x)) & (\Psi_{i+1,xx}(x), \Psi_{i,xx}(x)) \\ (\Psi_{i-1,xx}(x), \Psi_{i+1,xx}(x)) & (\Psi_{i,xx}(x), \Psi_{i+1,xx}(x)) & (\Psi_{i+1,xx}(x), \Psi_{i+1,xx}(x)) \end{pmatrix}, \\
D^{e_i} &= \begin{pmatrix} (\Psi_{i-1}(x), \Psi_{i-1,x}(x)) & (\Psi_i(x), \Psi_{i-1,x}(x)) & (\Psi_{i+1}(x), \Psi_{i-1,x}(x)) \\ (\Psi_{i-1}(x), \Psi_{i,x}(x)) & (\Psi_i(x), \Psi_{i,x}(x)) & (\Psi_{i+1}(x), \Psi_{i,x}(x)) \\ (\Psi_{i-1}(x), \Psi_{i+1,x}(x)) & (\Psi_i(x), \Psi_{i+1,x}(x)) & (\Psi_{i+1}(x), \Psi_{i+1,x}(x)) \end{pmatrix}.
\end{aligned} \tag{3.34}$$

Direct calculations of the inner products in the matrices in (3.34) give:

$$\begin{aligned}
A^{e_i} &= \frac{h}{30} \begin{pmatrix} 6 & 13 & 1 \\ 13 & 54 & 13 \\ 1 & 13 & 6 \end{pmatrix}, \quad B^{e_i} = \frac{2}{3h} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \\
C^{e_i} &= \frac{2}{h^2} \begin{pmatrix} -1 & 0 & 1 \\ 2 & 0 & -2 \\ -1 & 0 & 1 \end{pmatrix}, \quad D^{e_i} = \frac{1}{6} \begin{pmatrix} -3 & -8 & -1 \\ 2 & 0 & -2 \\ 1 & 8 & 3 \end{pmatrix}.
\end{aligned}$$

Combining together the contributions of all elements in (3.30)–(3.32), we get the assembled equations

$$AU^{n+1} + \frac{3i\tau}{4\epsilon}BU^{n+1} = -\frac{3i\tau}{4\epsilon}BU^{n*} + AU^{n*}, \tag{3.35}$$

$$A[\Phi^{n+1/2} + \Phi^{n-1/2} - 2f(U^n) - 2f(V^n)] = 0, \tag{3.36}$$

$$AV^{n+1} - \frac{\tau}{4}CV^{n+1} = \frac{\tau}{4}CV^n + \frac{\tau}{2}D\Phi^{n+1/2} + AV^n, \tag{3.37}$$

where $U^n, V^n, U^{n*}, \Phi^{n-1/2}, f(U^n), f(V^n)$ are M -dimensional vectors with the forms

$$\begin{aligned}
U^n &= (U_0^n, U_1^n, \dots, U_{M-1}^n)^\top, \quad V^n = (V_0^n, V_1^n, \dots, V_{M-1}^n)^\top, \\
U^{n*} &= (U_0^{n*}, U_1^{n*}, \dots, U_{M-1}^{n*})^\top, \quad \Phi^{n-1/2} = (\Phi_0^{n-1/2}, \Phi_1^{n-1/2}, \dots, \Phi_{M-1}^{n-1/2})^\top, \\
f(U^n) &= (f(U_0^n), f(U_1^n), \dots, f(U_{M-1}^n))^\top, \quad f(V^n) = (f(V_0^n), f(V_1^n), \dots, f(V_{M-1}^n))^\top,
\end{aligned}$$

and the matrix A, B, C and D are assembled by the element matrix $A^{e_i}, B^{e_i}, C^{e_i}$ and D^{e_i} , respectively.

To start the evolution of the vectors U^n and V^n , the value of U^0 and V^0 must be determined. To do so, we combine the formulation (3.19)–(3.21) and boundary condition $u_x(x_i, 0) = 0$, which lead to the relations $U_{-1}^0 - U_0^0 = 0$. The following matrix equation is obtained

$$\begin{pmatrix} 2 & 0 & 0 & \dots & 0 & 0 & 0 \\ 1 & 1 & 0 & \dots & 0 & 0 & 0 \\ 0 & 1 & 1 & \dots & 0 & 0 & 0 \\ \vdots & & \ddots & \ddots & & & \vdots \\ 0 & 0 & 0 & \dots & 1 & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} U_0^0 \\ U_1^0 \\ U_2^0 \\ \vdots \\ U_{M-1}^0 \\ U_M^0 \end{pmatrix} = \begin{pmatrix} u_h^0(x_0) \\ u_h^0(x_1) \\ u_h^0(x_2) \\ \vdots \\ u_h^0(x_{M-1}) \\ u_h^0(x_M) \end{pmatrix}. \tag{3.38}$$

The vector U^0 can be easily solved by the recursive algorithm from (3.38). Similar calculations can be done to get V^0 and $\Phi^{-1/2}$ by using the value of $u_h^0(x)$ and $v_h^0(x)$, where the initial value of ϕ is defined by $\phi^{-1/2} = |u^0|^2 + (v^0)^2$. Above all, we summarize the steps for solving (3.12)–(3.16) as:

Step1: Compute $\{U_i^0\}, \{V_i^0\}$ and $\{\Phi_i^{-1/2}\} (i = -1, 0, 1, \dots, M-1, M)$ by solving the matrix Eq. (3.38).

Step2: Compute $u_h^{0*}(x_i) (i = 0, 1, \dots, M)$ by applying the relationship $u_h^{0*}(x) = \exp(-\frac{i\tau}{2\epsilon}v_h^0(x))u_h^0(x)$. Then solve the Eq. (3.38) to get U^{0*} by replacing the vector of the right side of (3.38) with u_h^{0*} .

Step3: Solve the system of Eqs. (3.35)–(3.37) to get the value of U^1, V^1 and $\Phi^{1/2}$.

Step4: According to the relationship (3.19), $\{u_h^1(x_i)\}, \{v_h^1(x_i)\}, \{u_{hx}^1(x_i)\}, \{v_{hx}^1(x_i)\}$ and $\{\phi_h^{1/2}(x_i)\} (i = 0, 1, \dots, M-1, M)$ are obtained. This completes the computation of the value of the first level.

Step5: Repeat the process: Step2 → Step3 → Step4 until getting the $(n+1)$ th level value $\{u_h^{n+1}(x_i)\}$, $\{v_h^{n+1}(x_i)\}$, $\{u_{hx}^{n+1}(x_i)\}$ and $\{v_{hx}^{n+1}(x_i)\}$ ($i = 0, 1, \dots, M-1, M$). That completes the whole computing process.

4. Conserved properties

Theorem 1. Let u_h^{n+1} , v_h^{n+1} and $\phi_h^{n+1/2}$ be the solution of (3.12)–(3.16), then the following conserved properties hold:

$$I_1^* = \|u_h^{n+1}\|^2 = \|u_h^0\|^2, \quad (4.1)$$

$$I_2^* = \int_{\Omega} v_h^{n+1} dx = \int_{\Omega} v_h^0 dx. \quad (4.2)$$

Proof. Choosing $\varphi = u_h^{n+1} + u_h^{n*}$ in (3.13) and taking the real parts of it, we have

$$\|u_h^{n+1}\|^2 = \|u_h^{n*}\|^2. \quad (4.3)$$

Multiplying (3.12) by $\bar{u}_h^{n*}$ and integrating it on Ω , we get

$$\|u_h^{n*}\|^2 = \int_{\Omega} \exp\left(-\frac{i\tau}{2\epsilon} v_h^n\right) u_h^n \cdot \exp\left(\frac{i\tau}{2\epsilon} v_h^n\right) \bar{u}_h^n dx = \int_{\Omega} u_h^n \cdot \bar{u}_h^n dx = \|u_h^n\|^2. \quad (4.4)$$

It is easy to obtain I_1^* from (4.3) and (4.4).

Notice that the Eq. (3.15) is equivalent to

$$\frac{1}{\tau} (v_h^{n+1} - v_h^n, \varphi) + \frac{1}{4} (v_{hxx}^{n+1} + v_{hxx}^n, \varphi) + \frac{1}{2} (\phi_{hx}^{n+1/2}, \varphi) = 0. \quad (4.5)$$

Taking $\varphi = 1$ in (4.5) and applying the zero boundary conditions, we arrive at

$$\int_{\Omega} v_h^{n+1} dx = \int_{\Omega} v_h^n dx. \quad (4.6)$$

That completes the proof of the second conserved quantity. $\square$

The third quantity I_3 which presented in Section 2 will be calculated using a numerical quadrature.

5. Numerical experiments

In this section we choose the initial condition which obtained from [4],

$$u(x, 0) = -\frac{6}{5} \sqrt{3} \alpha \frac{1}{\cosh \xi} \tanh \xi \exp \left\{ i x \left(-\frac{\epsilon x}{3} \right) \right\}, \quad v(x, 0) = -\frac{9}{5} \alpha \frac{1}{\cosh^2 \xi},$$

where $\xi = (\alpha/10)^{1/2} x$, α and ϵ are constants. To compute the numerical solution, the following parameters are used $x_l = -30, x_r = 30, \alpha = 0.4, \epsilon = 1$.

The discrete conserved quantities are defined as follows:

$$I_1^n = \|u_h^n\|_l^2 = h \cdot \sum_{j=1}^M |u_{hj}^n|^2, \quad I_2^n = h \cdot \sum_{j=1}^M v_{hj}^n, \\ I_3^n = 3 \|u_{hx}^n\|_l^2 - \frac{1}{3} \|v_{hx}^n\|^3 + h \sum_{j=1}^M v_{hj}^n |u_{hj}^n|^2 + \frac{1}{3} \|v_{hx}^n\|_l^2.$$

Throughout the computation, the conservation errors are closely monitored, and they are defined by

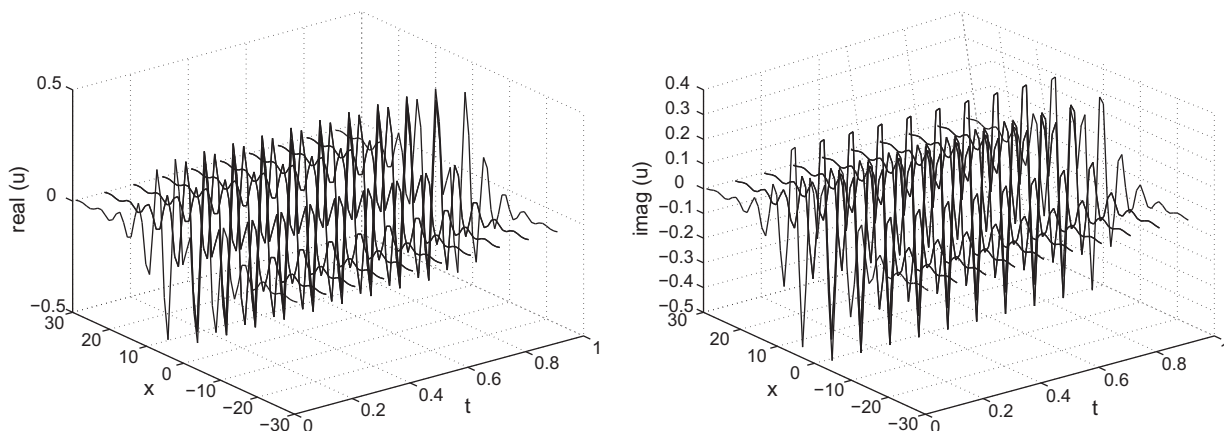
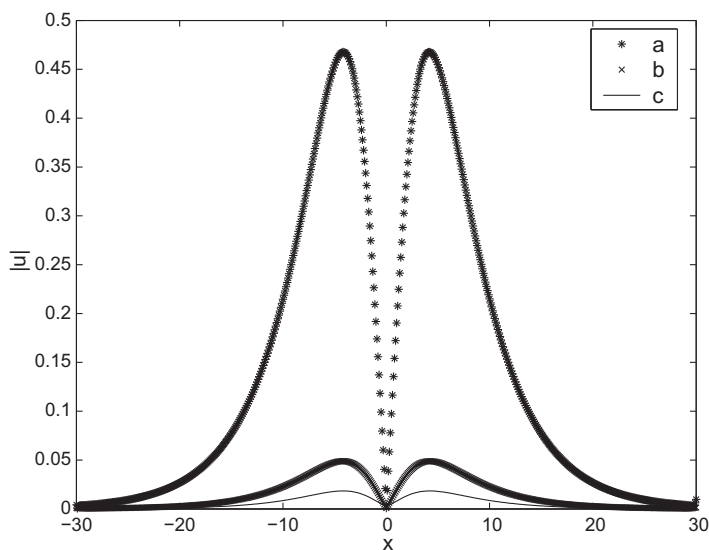
$$H_1 = \frac{|I_1^n - I_1^0|}{|I_1^0|}, \quad H_2 = \frac{|I_2^n - I_2^0|}{|I_2^0|}, \quad H_3 = \frac{|I_3^n - I_3^0|}{|I_3^0|}.$$

The I_1^0, I_2^0 and I_3^0 represent the conserved quantities at the initial time. For the case with known exact solutions, we define the solution maximum errors as $L_{\infty}^u = \max_j |u_{hj}^n - u(x_j, t_n)|$, $L_{\infty}^v = \max_j |v_{hj}^n - v(x_j, t_n)|$.

The three invariants obtained from the proposed scheme and the relative errors of them are shown in Table 1. It can be seen that our schemes conserve the conserved quantities well. In Fig. 2, the change of the real parts (left) and the imaginary parts (right) for u with $\epsilon = 8$ are displayed, which computed by our proposed scheme with $h = 0.1, \tau = 0.01$. The real part and the imaginary part of the solution u oscillate rapidly. Fig. 3 reflects the variation of the amplitude of $|u|$ with the change of parameter ϵ . The amplitude of $|u|$ decreases with parameter ϵ increasing. The value of the $|u|$ at each point varies slowly with time for $\epsilon = 8$, which may be due to the weaker nonlinearity for the bigger ϵ . Figs. 4 and 5 depict the convergence rate of the

Table 1Conserved quantities analysis for the system with $h = 0.1$ and $\tau = 0.1$ at different time.

Time	I_1	I_2	I_3	H_1	H_2	H_3
0.0	2.749183	-7.636708	-1.150562	–	–	–
2	2.749183	-7.636711	-1.155885	4.761754e-9	3.086207e-7	4.625811e-3
5	2.749183	-7.636715	-1.185460	9.114190e-9	8.776899e-7	3.033062e-2
8	2.749183	-7.636699	-1.200511	1.296860e-7	1.212599e-6	4.341209e-2

**Fig. 2.** Real parts (left) and imaginary parts (right) of u with $\epsilon = 8$.**Fig. 3.** The profile of $|u|$ with $h = 0.1$, $\tau = 0.01$ at $T = 1$ with different value of ϵ : a. $\epsilon = 1$, b. $\epsilon = 2$, c. $\epsilon = 8$.

numerical scheme with the different spatial step h and fixed $\tau = 0.0001$ at $T = 0.01$ and with the different temporal step τ and fixed $h = 0.5$ at $T = 0.05$, respectively. It can be found that the convergence rate increases with the decreasing of the other mesh size when we fix one of the mesh size. The convergence rate on the spatial direction is close to 3 for the function v in Fig. 4. In Fig. 5, the convergence rate on the temporal direction is close to 2. Fig. 6 shows the evolution of $|u|$ and v with $\epsilon = 1$ from $T = 0.0$ to $T = 2$.

Table 2 provided the results of the numerical solutions by the SSQBS FEM, the semi-discrete FEM [10] and the CN finite difference method [4], respectively. Although the CN scheme costs the least computing time, it's accuracy is lower than the other two the FEM. The reasons may be lie in two facts. On the one hand, the approximate solution obtained by the finite difference method is just the discrete nodal value, while that obtained by the FEM is continuous function at each time level.

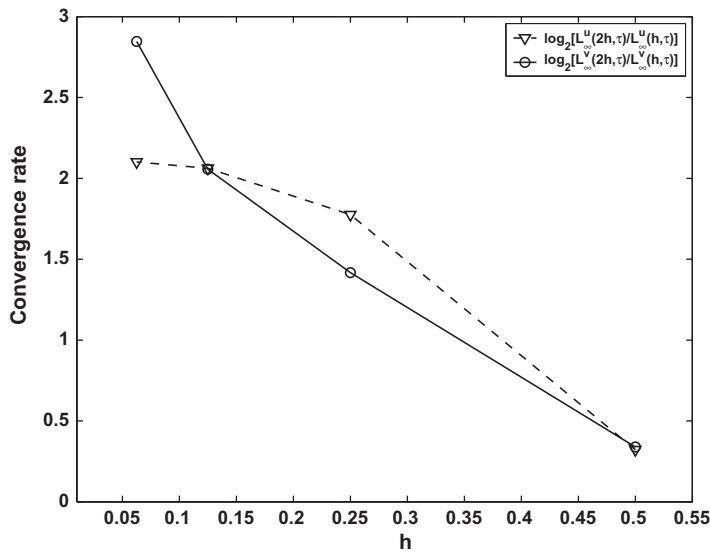


Fig. 4. The curve of the convergence rate calculated by the SSQBS FEM with the different h and the fixed τ .

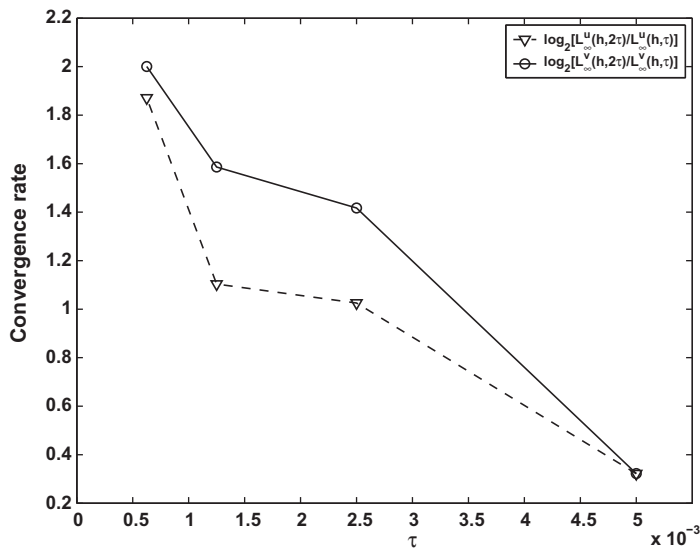


Fig. 5. The curve of the convergence rate calculated by the SSQBS FEM with the different τ and the fixed h .

It can be seen from the construction of the approximate solution (3.18) that the approximate solution are the linear combination of continuous base function on spatial direction. Thus, the precision of the finite difference method is lower than that of the FEM. On the other hand, the integral operation in the coefficient matrix (3.34) may cost much time, however, it is unnecessary in the finite difference method because the approximate solution obtained from finite difference method is discrete in spatial and temporal direction. For the two types of FEM, the SSQBS FEM has higher accuracy and costs less computing time than the semi-discrete FEM. We know that the CPU time depends essentially on the order of the finite element and the number of used mesh points. From the construction of the two types of elements point of view, the number of required mesh points in B -spline FEM for equal numerical precision is half of that in Lagrange FEM [24]. Meanwhile, the application of the split-step approach to Schrödinger-type equation and the introduction of a new viable in KdV equation can simplify the nonlinear terms in the Schrödinger-type equation and the KdV equation. Furthermore, the application of the relaxation approach to the new variable can avoid the complex iteration computing. Both of them save the computing time.

Moreover, we perform some tests of the accuracy with the cubic B -spline element to make a comparison. The numerical results are presented in the Table 2. Since the cubic B -spline base function has better smoothness than the quadratic B -spline

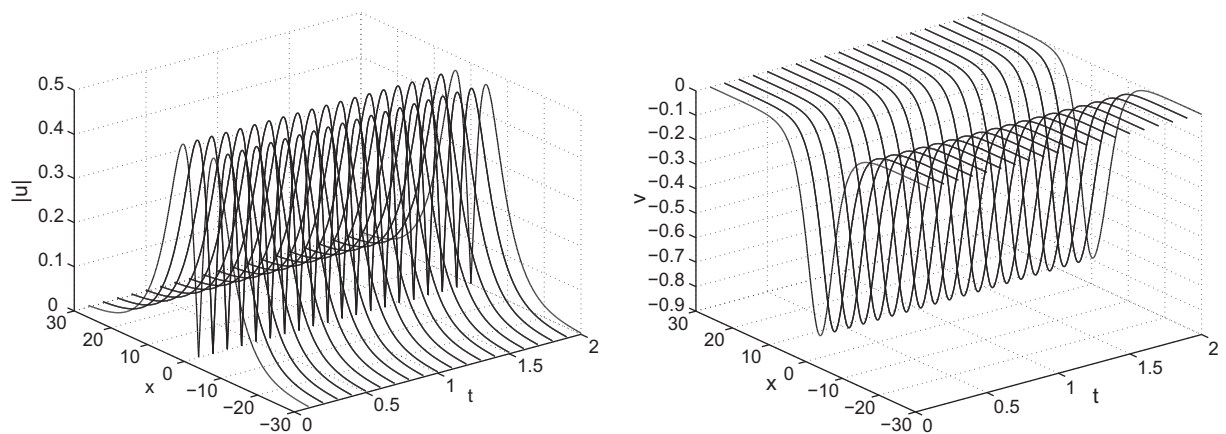


Fig. 6. The envelop of $|u|$ (left) and v (right) from $T = 0$ to $T = 2$.

Table 2

The Comparison of maximum errors using different schemes at $T = 0.1$ with $\tau = 0.0001$.

Method	h	L_{∞}^u	L_{∞}^v	CPU Time
SSQBS FEM	1	8.831056e-4	3.213056e-4	1.2030
	0.5	7.045855e-4	2.544081e-4	3.2350
SSCBS FEM	1	8.093879e-4	2.723093e-4	1.6375
	0.5	3.223085e-4	1.077306e-4	4.9220
Semi-discrete FEM	1	8.881089e-4	3.221997e-4	13.347
	0.5	7.210159e-4	2.559165e-4	25.204
CN	1	8.928318e-4	5.431211e-4	0.9840
	0.5	8.035485e-4	5.246808e-4	1.2810

base function at each finite element. Thus the former has higher accuracy than the latter when they approximate to the exact solution. Meanwhile, the cubic B -spline base function also adds the complexity of integration process and the bandwidth, which are both time-consuming procedures. Thus the split-step cubic B -spline finite element method (SSCBS FEM) costs more CPU time than the SSQBS FEM.

6. Conclusion

In this paper, we present a novel split-step quadratic B -spline finite element method for the coupled Schrödinger–KdV equations. The conserved properties of the full-discrete scheme are proved. Numerical experiments show that our scheme is efficient. Our method may be also used to solve other partial differential equations.

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